

Algebra 1
Unit 13
Lesson 9 Practice

Name Answer Key
Date _____
Period _____

To make a 14% acid solution, a chemist mixes 4 parts of 2% acid solution with an 18% acid solution. The expression $4(0.02x) + 0.18y$ represents the mixture, where x is the number of ounces of 2% solution and y is the number of ounces of the 18% solution.

Explain what each quantity means.

1. $(0.02x)$

Ounces of acid solution (2%).

2. $4(0.02x)$

Amount of acid solution in mixture (2%).

3. $0.18y$

Amount of acid solution (18%) in ounces.

Since 2018, the number of independently contracted food delivery drivers can be modeled by the equation $y = 1,675,000(1.064)^x$.

4. What does the x variable represent?

Years since 2018

5. What does the y variable represent?

Number of independently contracted food delivery drivers since 2018.

6. What does the quantity 1,675,000 represent?

Initial amount of independently contracted drivers.

7. What is the constant percent rate of change?

106.4% increase

The equation $y = -4(x - 2)^2 + 20$ models the height, y , in feet, of a ball, x seconds after it is thrown into the air.

Explain what each quantity represents.

8. $x - 2$

The number of seconds away the ball is before reaching maximum height.

9. 20

Maximum height of the ball, in feet.

Michael is buying coffee for \$5.75 per bag. He has a coupon for three free bags of coffee. He pays \$25.50 for a birthday cake. The expression $5.75(x - 3) + 25.50$ can be used to represent the overall cost.

10. What does the quantity $5.75(x - 3)$ represent?

The amount paid for the total amount of coffee, in bags, obtained.